

Metrics for eCommerce

What are top metrics you would track for an eCommerce website?

Things to Consider

Examples of top e-commerce websites include:

- US: Amazon, Wal-Mart, Apple
- India: Amazon, Flipkart, Snapdeal.
- China: Tmall, JD, Suning
- Europe: Tesco, Zolando, Otto

Consider sales, marketing and website metrics

Answer

Acquisition

Number of new customers

Cost per acquisition (CPA)

Cost per click (CPC)

Cost per impression (CPM)

Top search terms

Mailing list click through rate

Mailing list open rate

Mailing list conversion rate

Mobile app downloads

Activation

New registered users

Retention

Conversion rate

Recommendation engine conversion rate

Shopping cart abandonment

Shopping cart size

Monetization

Revenue per customer

Lapsed customers

Purchases per year

Revenue per click

Cost of sale (ad spend / revenue)

Lifetime value

Other

Cost of goods sold

Shipping time

Stockout

Market share

Margin

Metrics for Two-sided Marketplaces

What are top metrics you would track for a two-sided marketplace?

Things to Consider

Examples of two-sided marketplaces include:

Ridesharing: Uber, Lyft, Kuaidadi

Lodging: AirBnB, Booking.com, Expedia

Peer-to-peer marketplaces: Craigslist, eBay, OfferUp

Talent: Indeed.com

Answer

Buyer side

Acquisition

Mobile app downloads

Activation

Users with at least 1 search

Retention

Searches

Searches with 1+ Matches

CTR for search result

% satisfied transactions

Net promoter score

Monetization

Revenue

Avg. transaction size

of transactions

Seller side

Acquisition

Sellers

Seller growth rate

Activation

Sellers with at least 1 listing

Retention

Listings per seller

Net promoter score

Monetization

Gross marketplace volume

% fraudulent transactions

Other

Seller concentration, that is % of revenue generated by the top X% of sellers

Marketplace as a % of overall channel sales

Metrics for SaaS

What are top metrics you would track for a Software as a Service (SaaS) application?

Things to Consider

Examples of SaaS applications:

CRM: Salesforce

Messaging: Slack

Financial and human capital management: Workday

Answer

Acquisition

Leads

Virality

Activation

New registered users

Retention

Daily active usage

Time onsite

Interval between logins

Churn

Monetization

Conversions

Deals

Renewal rate

Monthly recurring revenue

Revenue per user

LTV

Other

Uptime

Renewal rate

Metrics for Mobile Apps

What are top metrics you would track for a mobile application?

Answer

Acquisition

of mobile installs

Cost per install

Activation

of accounts created, after mobile download

Retention

Daily and monthly active usage

Time in app

Star rating

Session length

% of users that rate the app

Monetization

% of paid users

Lifetime value

Average revenue per user

Churn

Metrics for Publishers

What are top metrics you would track for a publisher's website?

Things to Consider

Examples of publishers:

News: New York Times, India Times, Sina.com

Sports: ESPN

User-generated content: Reddit, Quora, Medium.com

Answer

Acquisition

Unique visitors per month

Sessions per month

Monthly page views

Activation

of registrations

Retention

Pages per visit

Time on site per visit

Monthly minutes on site

Monetization

Display ad rates (per thousand)

Banners per page

CPC ads per page

CPC (Cost per Click)

CTR (Click through rate)

Total CPC ads shown

Total Clicks

CPA ads per page

Total CPA ads

CPA (Cost per Acquisition)

CTR on CPA Ads

Total Clicks on CPA Ads

Conversion Rate (on CPA Ads)

Total Conversions

Average Sale

Total Sales

Affiliate Percentage

Metrics for User-generated Content Website

What are top metrics you would track for a user generated content website?

Things to Consider

Examples of user-generated content websites include Reddit, Quora and Medium.com

Answer

Acquisition

Visitors

Returning visitors

Activation

Registered users

Retention

Voters or Flaggers

Commenters

Posters

Moderators

Monetization

See items from the answer, Publisher Metrics

Metrics for Support Tickets

What are top support tickets metrics you would track?

Things to Consider

What are the business goals?

Which metrics correlate with saving money?

Which metrics correlate with increasing customer satisfaction?

Answer

of customer tickets

of customer tickets by type (email, call or chat)

Average resolution time

Concern classification

Net promoter score

First call resolution

Average # calls per resolution

Uber KPIs

Based upon what you know about Uber and its business model, what are some of the KPIs (key performance indicators) that you would want to focus on to judge how a market is doing overall?

Things to Consider

From a business goal perspective, what's more important to Uber: revenue or market share?

From a marketing perspective, what's most important: awareness, interest, trial or purchase?

What are the most important recurring usage metrics?

Common Mistakes

Not getting familiar with the topic beforehand, considering how widely Uber's business model and metrics are covered by mainstream business press.

Answer

CANDIDATE: Hmm, I can think of a few KPIs:

Number of rides per day. The more, the better.

Average miles of each ride. Are people actually taking rides for long distances or short distances? If it's mostly long distances, it might be a temporary thing. Short distances mean it's probably more stable.

Average rating across all Uber drivers. Are our drivers good in our customers' eyes?

Surge pricing as a % of total time. How often do customers have trouble finding an Uber ride? While drivers make more money when there's surge pricing, it turns off customers. We need to find a healthy balance.

Number of new drivers who get at least X amount of rides each time. Are we getting new drivers who are appealing the market? Are they finding decent work every day?

Most Important Metric: Two-Sided Marketplace

What is the most important metric for a two-sided marketplace like Uber?

Answer

CANDIDATE: Supply-side capacity is the most important metric for two-side marketplaces. For Uber, supply-side capacity is the number of drivers. For eBay, it's the number of seller listings. For AirBnB, it's the number of apartment rentals.

Take Uber for example. The more drivers there are, the more likely Uber can:

Cover a geographic location, especially an area where there's demand

Increase the likelihood of matches, improving customer satisfaction. Uber calls this minimizing zeroes.

Zeroes are situations where a passenger fires up the app and can't find a driver.

Minimizes customer wait times.

Minimize usage of surge pricing, improving number of completed transactions

Increase word-of-mouth recruiting for drivers, especially if current drivers tell potential drivers about the compensation possibilities

While it may feel that addressing customer demand could be as important, consider the following:

Uber riders need drivers more than the drivers need the riders. In other words, the drivers can find alternative and similar ways to make money, whereas the alternative to Uber rides, whether it's taxi (not enough coverage, poor service) or public transportation (inconvenient, unreliable) is significantly poorer.

If the riding experience is not of sufficient quality (largely controlled by the drivers), riders are unlikely to adopt Uber's service.

If the riding experience is of sufficient quality but unavailable (not enough drivers in a geographic area), then riders cannot adopt Uber's service.

Most Important Metric: eCommerce

Let's say you are a venture capitalist. What would be the most important metric you'd want to see if you're considering an investment?

Things to Consider

Venture capital is a high-risk and high-reward business. In other words, they're looking for opportunities where they can make billions of dollars.

For an eCommerce to be worth of billions of dollars, they will have to dominant the industry, like Amazon. By dominating, they can enjoy the economies of scale benefits such as super brand, logistics infrastructure and negotiation power over suppliers.

Gross merchandising volume represents the total sales volume, in dollars, flowing through an eCommerce site.

Common Mistakes

Giving a short, uninspiring response (e.g. "I'd go with profits.")

Answer

CANDIDATE: There are several eCommerce metrics to choose from including:

Conversion Rate	Customer acquisition cost	Site traffic by source
Average order size	Margins	Gross merchandising volume
Shopping cart abandonment	Site traffic	Newsletter subscribers

If I were to choose one metric, I'd choose gross merchandising volume. Here's why:

As a venture capitalist, I'd like to know how big the business could be. Gross merchandising volume (GMV) can give me a sense of how the large the business might be.

From GMV, we can compute the revenue, whether the eCommerce site is based on extracting a commission like eBay or deriving gross margins by selling goods like Amazon.

GMV is also driven by important metrics such as traffic, conversion rate and average order size.

GMV is also a proxy for market position.

As much as I like the GMV metric, there are a few drawbacks:

GMV does not factor in customer acquisition cost.

GMV does not consider cost of sales or other margin information.

GMV may not be correlated with cash flow.

Drop in Hits

There was an 8% drop in hits to Google.com. Larry Page walks into your office and asks you to think and list what the reasons might be.

Things to Consider

- Seasonality
- Server maintenance
- Changes in referrers
- Changes in website layout
- Changes in website content
- Changes in website loading times
- Software bugs

Common Mistakes

- Having a short, unsatisfying list of metrics.

Answer

CANDIDATE: I would first begin by examining if this is country or region specific. Maybe there is censorship happening or a country has problem with its network infrastructure, or maybe our servers for that area is down. If that's not the case, I would see if this is desktop vs. mobile. I would continue to check if it's platform specific (e.g. iOS, Android, Amazon, Windows Phone) if it's mobile or browser specific (e.g. Chrome, Edge, Safari, Opera) or even if it's OS specific (Windows, Mac, and Linux).

Since this is an 8% drop in hits, I would also check with our tracking. Maybe our tracking has gone awry. I don't think I need to ask about other technical problems (like search not working on specific terms) because if we have them, I am sure Google has all the necessary checks in place to detect this and we would be facing a different question.

I would then begin to see if returning users or new users are the culprit. If returning users, perhaps a recent UI redesign is causing people to dislike our page. Maybe our search algorithm has a problem and the results are not as good as before. Or perhaps we added annoying ads or our account system is doing something weird. These are all likely things that would cause recurring users to leave.

If it is not the returning users, it must be our new users. We can dissect this through direct or referral. For direct, as I said earlier, if google.com is down we would have known by now so I don't think that's the problem. We can look at referral. First, is our marketing suffering from problems? Maybe our budget was slashed, or the rate has gone up but we haven't changed it so our prices are no longer competitive. Or perhaps our targeting is off. Are we targeting people who have access to Internet and a computer?

Next, we can look at other sources of referral. Are the links to google.com still working for our other products? If there is a problem with that, that could be it. Or, if new users are getting to our page fine, we must dissect our

user experience. Maybe we're driving new users away with a different UI, different search algorithm or perhaps annoying pop-up ads.

This is the process I would go through. Is there any specific ones you want me to focus on?

INTERVIEWER: You've got a great list. Let's move on to the next question.

Amazon Web Services Metrics in Sydney

Imagine you are the AWS PM in Sydney. What are the top three metrics you'd look at?

Things to Consider

For those who aren't familiar with AWS, it helps to first brainstorm a list of potential metrics to make sure nothing is missed.

From there, choose what you believe to be the top three metrics.

Be prepared to defend your positioning with clear rationale.

Common Mistakes

Blurting out three metrics and then changing one's top three after they realize something better. It comes across as wishy-washy.

Answer

CANDIDATE: Interesting question. Could I brainstorm for a bit?

Candidate takes 30 seconds.

CANDIDATE: I would look at these three metrics:

Profit. I want to know how much profit our service in Sydney is making us. This is important, because we need to pour more resources into maintaining and expanding this. If we are not making profit, we need to think about why this market has more room to grow.

Companies that need similar services but are not using AWS. I want to know why they are not using AWS. Are there local or global competitors? Are they using them over us because of pricing or maybe different policies or customer service? This would also give us the potential market we could get. If this market is very small and we are not making profit, then we need to rethink Sydney.

Companies that promote products outside of Sydney who are using the servers in Sydney. How many companies are using our Sydney infrastructure who are promoting products elsewhere? If there are a lot, why are they not using products in these other areas? Do we have services in those areas? Why not? This would give us an idea on whether or not we should expand to nearby areas.

Success Metric for UberPool

What should be the main success metric for UberPool?

Things to Consider

- # of UberPool requests
- # of UberPool matches
- # of UberPool trips, given a time period
- # of active UberPool users
- Reduction in city congestion, due to UberPool usage

Common Mistakes

Choosing vanity metrics such as visitors and visits to the www.uber.com/ride/uberpool/ page

Answer

CANDIDATE: Hmm, UberPool is the one where you are carpooling with multiple people right? For one ride? And each rider is being picked up from different locations and going to different pickups?

INTERVIEWER: Yes.

CANDIDATE: Hmm, I would track these:

- Average number of people per UberPool. A pickup location might be two people instead of one.
- Average number of stops per UberPool.
- Percentage of UberPool compared to other Uber rides.
- Number of times people thought about UberPool, then gave up and chose something else.
- Average rating differences between UberPool experience vs. other Uber rides.
- Average number of UberPool trips per month by repeated users.

The main success metric is probably the percentage of UberPool compared to other Uber rides.

INTERVIEWER: You mentioned percentage earlier. What percentage do you think is healthy for UberPool compared to other Uber rides?

CANDIDATE: Hmm, I would look at historical data and see what data we were happy with before. If we don't have that and I have to guess within 10 seconds, probably around 10%.

INTERVIEWER: That's low. Why 10%?

CANDIDATE: Well, most people who take Uber do it for the convenience, but also a large sector of people use Uber because they need to get somewhere fast. Speed is of the essence here, so I don't think it'll be that popular. Sure, plenty of people want to save, but I don't think it'll be that high.

Metrics for Uber Pick-up

What metric would you measure if we were focused on improving the Uber pick-up experience?

Things to Consider

- Number of people opening the request page
- Number of people making a ride request
- Number of requests that are fulfilled
- Time to fulfill request
- Post-trip satisfaction

Common Mistakes

- Focusing too much on low value-add items such as whether the driver offers water to its riders or whether the driver is a good conversationalist.

Answer

CANDIDATE: When you mean pick-up, do you mean just the waiting period, or do you count the finding a driver part of the experience?

INTERVIEWER: I mean the whole experience.

CANDIDATE: Okay, I would track the following:

- How long does it take someone to get a ride?
- How long does it take the driver to arrive at the pickup location?
- Average rating of the drivers.
- How long the phone call is between the driver and the rider? I'm guessing shorter phone calls means a better rider and driver experience. This could be misleading though, so we need to look at this in conjunction with the other metrics.
- How many drivers do users look at before picking one?
- How many times surge pricing happened? Frequent price surges indicate higher rider fares and a shortage of drivers.

Go-To-Market and Success

How would you measure the success of a go-to-market strategy?

Things to Consider

As a starting point, think about all the key pieces of the marketing funnel: awareness, interest, trial and purchase.

Common Mistakes

Setting the wrong goals.

Not setting sufficiently high goals.

Not having the data or getting incorrect data, leading the team to make the wrong conclusions.

Answer

CANDIDATE: Hmm, this is an open-ended question. Let me first describe what a go-to-market is, and then I will ask some clarifying questions.

Go-to-market can probably be best explained as a marketing process. When you start, you should think about your customers. Who are they? Are we targeting consumers or businesses? What kind of consumers or businesses? Once we identify a target segment, we can think about what marketing campaigns can target this segment most effectively. For example, if we are targeting businesses, a direct sales effort is more effective, especially since business products are a high involvement process. Then we need to think about how they can purchase our service. Going back to the business example, they can sign contracts with us, or they can sign for our service on their own, using a credit card like a software-as-a-service business like Salesforce. Our account managers can then serve as first point of contact should our business customers encounter any problems.

INTERVIEWER: Thanks for the go-to-market explanation. So what success metrics would you look at?

CANDIDATE: I would measure various metrics:

Reach. How many customers did we reach out to?

Conversion rate. Of those that we contacted, how many purchased our service or product?

Customer acquisition cost. How much are we spending to market our products and goods? I'd like to compare this by product line, geography and sales channel.

Revenue and profit. Most tech companies care about revenue, especially since tech companies have significant fixed costs such as R&D labor that can't be easily assigned to a single P&L. However, it would be best if we can track profitability. Numerous companies have incorrectly sustained poor performing businesses because they allocated costs incorrectly.

Market share. Market share is most critical for winner-take-all markets such as two-sided marketplaces like AirBnB, Uber and eBay.

Word of mouth. How many customers come to us because of other customers? We can track this using the industry measure for virality, the K-factor.

